

SUMMARY

Recent Computer Science graduate from Michigan Technological University with hands-on experience across software engineering, applied machine learning, and formal methods research. Seeking an entry-level software engineering role where I can apply my background in leadership, project management, and full-stack development.

EDUCATION

Michigan Technological University | Houghton, MI

BS Computer Science

4.0 GPA - April 2026

MS Computer Science

April 2027

PROFESSIONAL EXPERIENCE

GE HealthCare | Edison Engineering Development Program (EEDP) Intern

STO Smart Devices Team — Automatic Mask Generation with Synthetic Medical Images

May 2026 – Present

- Developed a latent diffusion model (LDM) to jointly generate synthetic medical images and corresponding segmentation masks via cross-attention-based automatic mask generation, extending prior OVAM-based work.
- Used a BERT-based text encoder to condition image and mask generation on clinical descriptions.
- Achieved up to 0.89 accuracy on mask generation, with particularly strong performance on lung segmentation.
- Preparing a paper disclosure submission based on this work.

CT Scanner Desktop Team — AI Customer Support Aggregation Tool

May 2025 – Dec 2025

- Researched and implemented machine learning models for semantic embedding of customer support issues.
- Identified and tested NLP models achieving up to 0.65 recall for identifying relevant topics in a document retrieval application.
- Designed and developed a responsive Angular application with OpenAPI code generation through API-driven development.
- Saved an estimated 3000 developer-hours annually through the improved customer support workflow.
- Extended internship through the fall to write functional and integration test code for the CT Scanner Desktop application, strengthening release reliability.

Phase 1 Engineering | Product Development and Research Intern

Simulation Enhancement Tools

Apr 2024 – Aug 2024

- Created an internal visualization tool in Python to extend the capabilities of simulation software for direct, parameterized comparison of design outputs.
- Built a backend in C for the tool's I/O functions, achieving a 3x increase in processing speed and improving the overall user experience.

Reduced Cost Haptic Device

Jun 2023 – Aug 2023

- Collaborated with senior engineers to reverse-engineer a market solution, then designed primitive amplifier circuitry that removed the need for a dedicated amplifier, reducing total device cost by 50%.
- Prototyped solenoid alternatives using EM simulation and 3D printing.

RESEARCH

Rules as Code Researcher | Michigan Technological University

Oct 2024 – Present

- Researched formal methods for interpreting real-world rules using computational models, and developed an Alloy Analyzer model of student registration and graduation requirements.
- Built a full-stack application with a Spring Boot Java backend, Angular frontend, and MongoDB database to support the research.
- Awarded Summer Undergraduate Research Fellowship (SURF) and Undergraduate Research Internship Program (URIP) funding.

LEADERSHIP AND INVOLVEMENT

President | Innovative Global Solutions

Jan 2026 – Present

- Lead a 30+ member student engineering enterprise that builds solutions for real-world clients, overseeing day-to-day operations and driving continuous improvement.
- Coordinate project needs and resource allocation between teams and project leaders across the enterprise.

Project Lead | Infant Incubator, Innovative Global Solutions

Jan 2025 – Dec 2025

- Led a team to receive the "Innovation Award" at the Spring 2025 Design Expo.
- Managed project tracking and deadlines across subsystems and subteams.

SKILLS

Technical: Java, Python, C, R, SAS, TypeScript, SQL, Spring Boot, OpenAPI, Docker, Alloy Specification, MongoDB, MySQL, Git/GitHub, PyTorch, TensorFlow, Jenkins, Unit/Integration Testing

Professional: Project Management, Leadership, MS Office